CS146S Aut26 Final Project Proposal

Problem Statement

Early-career employees starting their first salaried job face many high-stakes financial decisions in their first month, usually with no training. They are required to fill out a W-4, choose a health plan, set a 401(k) contribution, and figure out what they can safely spend.

These are consequential decisions. Budgeting apps ignore personalized benefits and taxes and financial advisors are too expensive on a first salary. Online articles are not personalized to the reader's own numbers.

We will attempt to solve this problem by creating an app that uses AI to guide a new employee through their first-paycheck decisions and explaining each one with their own numbers, employee contracts, and healthcare plans.

Target User

Our target user is an early-career employee, recently starting their first salaried job with benefits. Every feature is designed around this person's first 30 days on the job, and around the yearly checkpoints that follow.

- **Context:** an offer letter or first pay stub, a benefits enrollment deadline, a bank account with a few months of history, and little financial training.
- **Goals:** make choices aligned to the user’s goals before deadlines pass, know what they can safely spend, and start building security.
- **Needs:** answers tied to their own salary, benefits, and spending, with explanations that are cited from reliable sources.

Initial Business or Use Case Context

Setting. In their first weeks at a new job, the user uploads a pay stub or offer letter and a bank CSV, then completes a guided setup in one or two sittings. They return at yearly checkpoints such as at open enrollment, tax season, and life events such as a raise.

Value. Every recommendation explains the concept, applies it to the user's own numbers, and shows what different decisions might alter their financial future. Users leave with a setup plan and the understanding to make the next decision themselves.

Business context. Free for individuals, or sold to employers as a financial-wellness benefit for new hires. The yearly decision calendar drives return visits.

Planned Features

One guided flow in which each step feeds the next. Financial math runs in deterministic, tested code; the LLM handles extraction, categorization, and explanations.

MVP (must-have)

1. **Paycheck decoder:** extract and explain each line of a pay stub or offer letter.
2. **Spending map:** auto-categorize a bank CSV (user-correctable) and compute monthly essentials.
3. **Emergency fund planner:** size a 3–6 month target and show time-to-goal at different savings rates.
4. **401(k) match optimizer:** show the contribution that captures the full match and the dollars gained or lost.
5. **"Why" explanations:** the concept, the user's numbers, and what would change the answer, grounded in IRS and plan sources.
6. **Saved profile and setup plan:** persist decisions and reasoning, and export a one-page plan.
7. **Health plan comparison:** HSA-eligible high-deductible plan vs. PPO, with expected cost and break-even point.
8. **W-4 walkthrough:** guided form completion with a projected refund or balance due.

Stretch (nice-to-have)

- **Savings priority ladder:** Roth IRA vs. 401(k) guidance with what-if toggles.
- **Annual check-ins and bank sync:** open enrollment and tax-season reviews (seeded data), and Plaid sandbox.

AI Tool Usage Plan

- **Code generation and refactoring:** Claude Code and/or Cursor scaffold the app and build parsers and calculators from specs we write first.
- **Automated testing:** AI-generated unit and property-based tests for every calculator, plus an eval set of labeled pay stubs and transactions measuring extraction and categorization accuracy.
- **Code review and QA:** Implemented for every PR
- **UI/UX prototyping:** AI-generated clickable mockups of the setup flow
- **CI/CD and deployment:** AI-written GitHub Actions run lint, tests, and evals on each PR, block merges on eval regressions.

We will log where AI helped, where it failed, and how we caught it, for the final reflection.